

Milestone 1: Technical Understanding and Research Framing

Loop Unrolling and Controller Delay in High-Level Synthesis

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I. MILESTONE 1: TECHNICAL UNDERSTANDING AND RESEARCH FRAMING

A. Topic

- The topic of this seminar paper is the impact of loop unrolling on controller delay in high-level synthesis.
- The main reference paper studies how different loop unrolling factors influence the delay of the controller generated during high-level synthesis.
- The paper focuses on the problem that loop unrolling can improve performance, but can also increase the complexity and delay of the finite-state machine controller.

B. Problem Statement

- Loop unrolling is a common optimization technique.
- It duplicates the body of a loop so that multiple loop iterations are executed together.
- In software compilation, loop unrolling is usually used to reduce loop overhead and expose more instruction-level parallelism.
- In high-level synthesis, loop unrolling has additional hardware effects.
- The unrolled loop body becomes part of the generated hardware structure.
- This can increase the number of operations, control states, multiplexers, and resource-sharing decisions.
- Because of this, the controller delay can increase.
- If controller delay is not considered, the synthesized design may fail to meet the required clock period.
- Therefore, the problem is not only how to reduce latency, but how to choose an unroll factor that improves performance while still satisfying timing constraints.

C. Motivation

- High-level synthesis allows designers to create hardware from high-level descriptions such as C or C++.
- This makes hardware development easier, but it also means that compiler-style optimizations need to be understood from a hardware perspective.

- A transformation that looks beneficial at the algorithmic level may create timing or resource problems after synthesis.
- Loop unrolling is a good example of this issue.
- A larger unroll factor can reduce the number of loop iterations.
- However, it can also increase the size and delay of the controller.
- This makes it important to study the tradeoff between latency reduction and controller-delay increase.

D. Core Technical Idea

- The main idea is to predict how the controller delay changes when the loop unroll factor changes.
- Instead of synthesizing every possible unroll factor, the paper proposes an estimation-based method.
- The method uses high-level information such as:
 - number of operations,
 - number of control steps,
 - number of FSM states,
 - number of state bits,
 - number of functional-unit control bits,
 - number of variables.
- These values are used to estimate FSM/controller delay.
- The estimated delay is then used to guide the selection of a suitable unroll factor.
- The goal is to find an unroll factor that improves latency while remaining inside a given delay budget.

E. Important Technical Concepts

- High-Level Synthesis:
 - Converts a behavioral description into RTL hardware.
 - Generates a datapath and a controller.
- Datapath:
 - Contains arithmetic units, registers, multiplexers, and memories.
- Controller:
 - Usually implemented as a finite-state machine.

- Controls when operations happen and which resources are selected.
- Loop Unrolling:
 - Duplicates loop body operations.
 - Reduces loop-control overhead.
 - Exposes more parallelism.
 - Can increase hardware complexity.
- Controller Delay:
 - Delay caused by FSM logic and control decisions.
 - Can become part of the critical path.
 - Can limit the achievable clock period.

F. Formalism and Model

- The paper uses an estimation model for FSM delay.
- A simplified form of the model is:

$$Delay = A \cdot \log(\#operations) \cdot (\#state\ bits) \quad (1)$$

- The constant A depends on hardware-specific factors such as functional-unit control bits and number of variables.
- The number of operations increases when the loop body is duplicated.
- The number of state bits depends on the number of FSM states or control steps.
- The unroll factor affects both operation count and controller complexity.

G. Assumptions

- The method assumes that the loop structure is suitable for unrolling.
- The loop bounds and unroll factors must be known or estimable.
- The delay model depends on the synthesis flow and target technology.
- The method assumes that controller delay can be estimated from high-level information.
- The paper focuses on controller delay, not the complete timing behavior of all possible hardware targets.

H. Scope Limitations

- The original experiments were performed using a specific behavioral synthesis framework and ASIC library.
- The exact numerical delay values may not directly transfer to modern FPGA tools.
- Modern HLS tools may apply additional optimizations that interact with loop unrolling.
- Controller delay may not always be directly visible in student-level HLS reports.
- In a student implementation, resource usage, latency, clock estimate, and timing slack may need to be used as indirect indicators.

I. Resource, Runtime, and Scalability Aspects

- Exhaustively synthesizing every possible unroll factor can be very time-consuming.
- The number of possible design choices grows when multiple loops are considered.
- Large unroll factors can increase:
 - operation count,
 - FSM complexity,
 - multiplexer size,
 - functional-unit sharing complexity,
 - synthesis time,
 - hardware area.
- The paper proposes estimation and pruning to avoid unnecessary synthesis runs.
- This makes the approach more scalable than exhaustive exploration.

J. Open Questions

- How accurately can the original delay model be reproduced using modern FPGA-based HLS tools?
- Can controller delay be separated clearly from datapath and routing delay in student-level synthesis reports?
- How does loop unrolling interact with loop pipelining in modern HLS tools?
- What is the best low-budget way to demonstrate the tradeoff experimentally?
- Would a Python simulation be sufficient, or should a small HLS experiment also be included?

REFERENCES

- [1] S. Kurra, N. K. Singh, and P. R. Panda, "The Impact of Loop Unrolling on Controller Delay in High Level Synthesis," DATE, 2007.